COMP 5331 (12 Sept)

In-class Ex. 1

threshold = 2

Q1

item	count
A	4
B	0
C	3
D	3
E	3

$L_1 = \{ [A], [C], [D], [E] \}$

Large 2-itemset Generation

Candidate Generation

Join / Prune step:

$C_2 = \{ (A,C), (A,D), (A,E), (C,D), (C,E), (D,E) \}$ (counts: 2, 3, 3, 2, 2, 2)

Counting step:

$L_2 = \{ (A,C), (A,D), (A,E), (C,E), (D,E) \}$

Large 3-itemset Generation

Candidate Generation

Join step:

$C_3 = \{ [A,C,D], [A,C,E], [A,D,E] \}$

Prune step:

$C_3 = \{ \cancel{[A,C,D]}, [A,C,E], [A,D,E] \}$ ([A,D,E] count: 2)

Counting step:

$L_3 = \{ [A,C,E], [A,D,E] \}$

Large 4-itemset Generation

Candidate Generation

Join Step:

$C_4 = ($)

Large itemsets — set union

$= L_1 \cup L_2 \cup L_3$

$= \{(A), (C), (D), (E),$

$(A, C), (A, D), (A, E),$

$(C, E), (D, E),$

$(A, C, E), (A, D, E))$

COMP5331 Knowledge Discovery in Databases (Fall Semester 2026)
Exercise 1
Association Rule Mining

Q1 Apriori Algorithm

Given the following transactions, and the support threshold = 2.

A	B	C	D	E
1	0	0	1	0
1	0	0	1	1
0	0	1	0	0
1	0	1	1	1
1	0	1	0	1

Follow the steps of the Apriori Algorithm and deduce L_1, C_2, L_2, C_3, L_3, C_4, … until all the large itemsets are discovered.

e.g. 1 {A, B}

prefix

e.g. 2 {A, B, C}

prefix

)Approch 1

if these two itemsets have the same prefix, we will generate the candidate

Prune Step

{A, C, D} — ① {A, C} ✓
— ② {A, D} ✓
— ③ {C, D} ✗

① to enumerate all possible 3 (4 3 1) 2-itemsets

② if one of these 3 (4 2 1) 2-itemsets is NOT large, remove this 3 (4 3) 3-itemset

{A, C, E} — ① (A, C) ✓
— ② (A, E) ✓
— ③ (C, E) ✓

{A, D, E} — ① (A, D) ✓
— ② (A, E) ✓
— ③ (D, E) ✓

~~Large~~

$L_3 = \{ \{A, B, C\}, \{A, B, E\} \}$

Large 4-itemset Generation

Candidate Generation.

Join Step:

$C_4 = \{ \{A, B, C, E\} \}$

Prune Step:

$C_4 = \{ \{A, B, C, E\} \}$

Prune Step

(A, B, C, E)

① (A, B, C) ✓

② (A, B, E) ✓

③ (A, C, E) ✗

④ (B, C, E) ✗

Your Approch

2-itemsets

if these two itemsets have a common item, we will generate the candidate

$L_2 = \{ (A, E), (C, E) \}$

My Approach

Join Step

$C_3^{(my)} = ($ $)$

Prune Step

$C_3'^{(my)} = ($ $)$

Your Approach

Join Step

$C_3^{(your)} = \{ \{A, C, E\} \}$ ① [A, C) ✗ ② [A, E) ✓ ③ (C, E) ✓

{A, C, E} ✓ ① (A, C) ② (A, E) ③ (C, E)

Prune Step

$C_3'^{(your)} = \{$ $\}$ $\{ [A, C, E) \}$

Claim 1 $C_3^{(my)} \subseteq C_3^{(your)}$

Claim 2 $C_3'^{(my)} = C_3'^{(your)}$

header table

horizontal link

depth / height of the tree

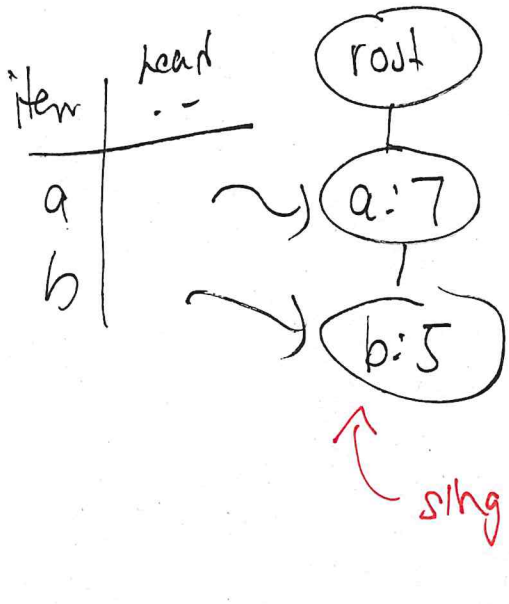
⇒ max. # of freq. items in a trans.

[a, b, c, e, f, g]: 1

[a, b, d]: 1

threshold = 3

Cond. FP-tree on "g" [14]



Step 1 : {g}:14

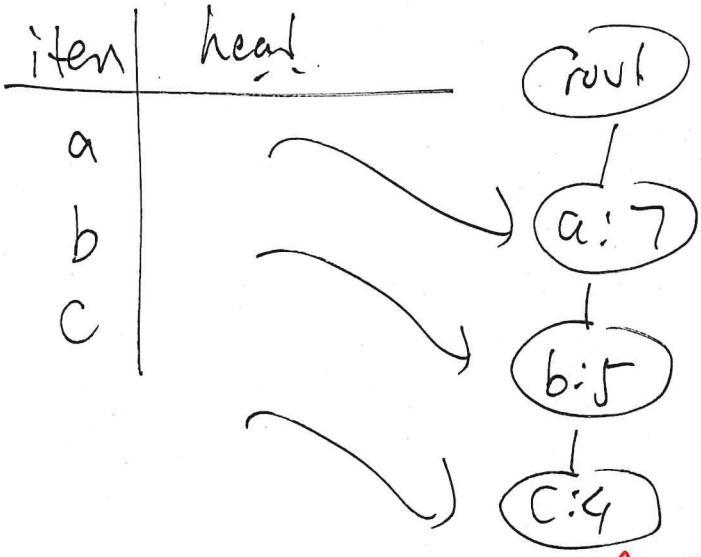
Step 2 :

(a): to enumerate all poss. itemsets from {a, b}

(b): to append item "g" to each poss. itemset

size	itemset	supp
1	{a, g}	7
	{b, g}	5
2	{a, b, g}	5

Cond. FP-tree on "g" [14]



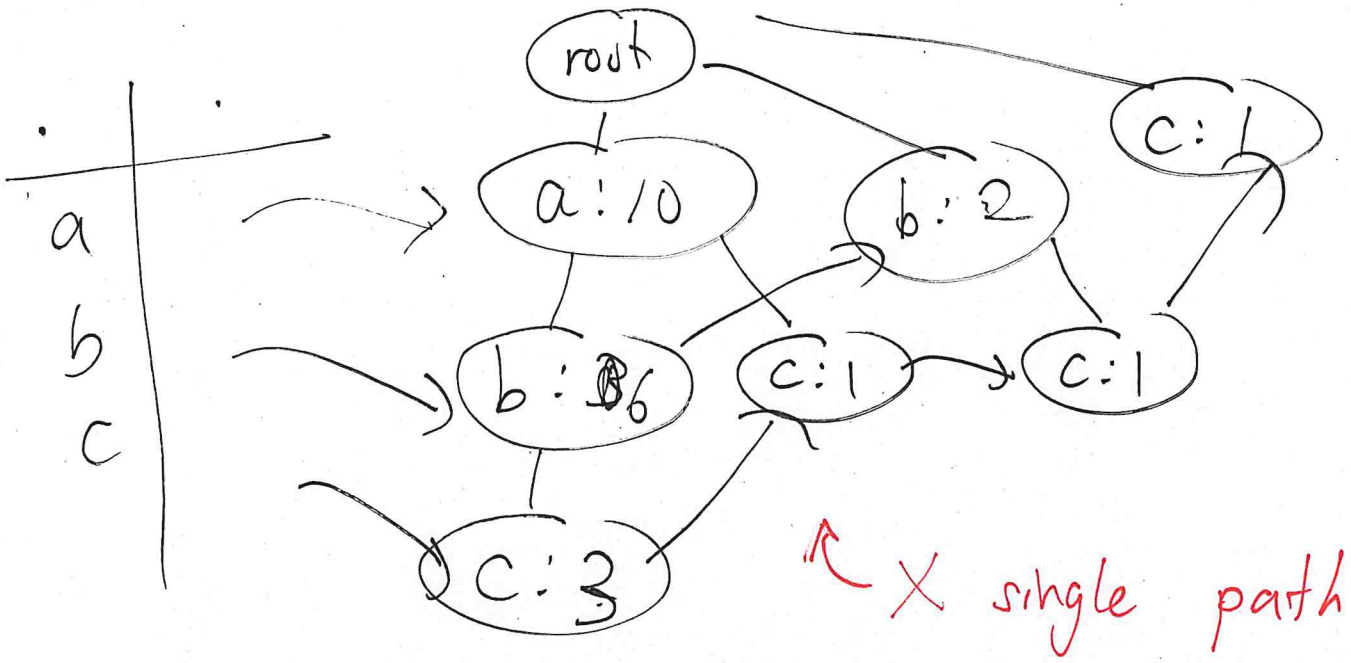
step1 [g]: 14

step 2.

↑ single path

size	items +	supp
1	{a, g}	7
	{b, g}	5
	{c, g}	4
2	{a, b, g}	5
	{a, c, g}	4
	{b, c, g}	4
3	{a, b, c, g}	4

Cond. FP-tree on "g" [4]



Cond. FP-tree on {c.o.9} ⑤ 6 threshold=3

① {a, b, c}: 3
{a, c}: 1
{b, c}: 1
{c}: 1

⇒

⑥ ~~{c, a, b}~~: 3
~~{c, a}~~: 1
~~{c, b}~~: 1
~~{c}~~: 1

②

item	freq.
a	4
b	4
c	6

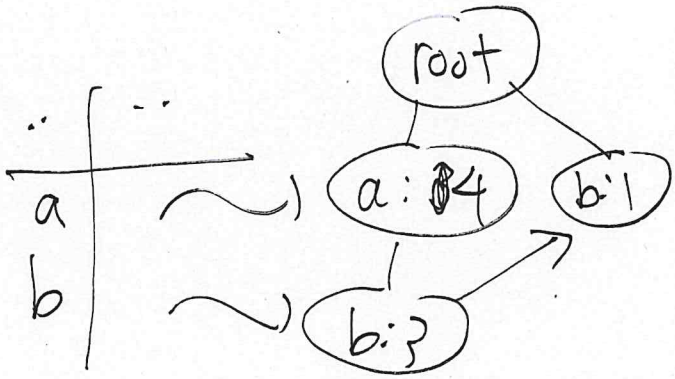
③

item	freq.
a	4
b	4
c	6

④

item	freq
c	6
a	4
b	4

⑦



Cond. FP-tre on ⟨ b, c, g ⟩ (5) 4

(1) {a, b}: 3

[b]: 1

(2)

item	freq
a	3
b	4

(3)

a	3
b	4

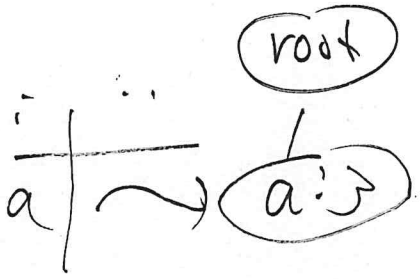
(4)

b	4
a	3

(6) ~~{b, a}: 3~~

{b}: 1

(7)



Cond Fp-2 on (a, c, g)

Cond. FP-tree or [b, g)

Cond. FP-tree on (a, g)